MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE COAXIAL, 50 OHMS
UNARMORED M17/065-RG165, AND ARMORED, M17/065-RG166.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cables described herein shall consist of this document and the latest issue of Specification MIL-C-17.

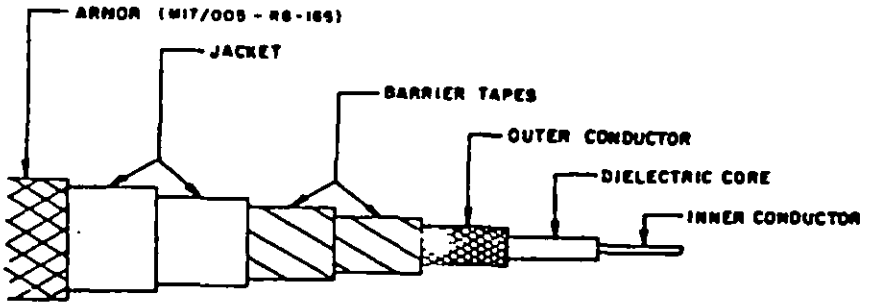


FIGURE 1. Configuration.

TABLE I. DESCRIPTION.

Components	Construction details
Inner conductor	Seven strands of silver-coated copper wire each strand, 0.0315 inch diameter Overall diameter: 0.094 inch ±0.001
Dielectric core	Type F-1: Solid, extruded PTFE. Diameter: 0.285 inch ±0.005.
Outer conductor	Single braid of AWG #34 silver-coated copper wire. Diameter: 0.340 inch maximum. Coverage : 93.2% nominal. Carriers : 24 Ends : 6 Picks/inch : 16.6 ±10%
Barrier tapes	Type FF-2: Two wraps of PTFE type, 0.005 inch thick each, by 1-inch with an approximate lay of 24 turns per foot and with 1/2 inch lap.
Jacket	Type V: Double braid of fiberglass. Diameter: 0.410 inch ±0.010.
Armor (M17/065-RG166)	Single braid of aluminum-alloy wire. Diameter: 0.470 inch maximum.

FSC 6145

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ENGINEERING INFORMATION:

Capacitance: 29.3 pF per foot, nominal.
Continuous working voltage: 3,700 Vrms, maximum.
Operating frequency: 3 GHz, maximum.
Velocity of propagation: 69.5 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -55° to +250°C.
Weight: M17/065-RG165 - 0.142 pound per foot, nominal.
M17/065-RG166 - 0.189 pound per foot, nominal.
Inner conductor properties:
DC resistance (maximum at 20°C): 0.170 ohms per 100 feet.
Elongation: 25 percent, minimum.
Engineering notes: This cable useful in general purpose high temperature applications. (M17/065-RG165, see connector series "N", "C", and "SC" per MIL-C-39012. NATO preferred type NWR-10.) (M17/065-RG166, see connector series "N", "C", per MIL-C-39012, "HN" per MIL-C-3650.)

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Visual and mechanical examination:
Eccentricity: 10 percent, maximum.
Adhesion of conductors:
Inner conductor to core: 4 pounds, minimum; 30 pounds, maximum.
Aging stability: +250° +5°C.
Dimensional stability: +250° +5°C.
Inner conductor from core: 0.312 inch, maximum.
Inner conductor from jacket: 0.125 inch, maximum.
Flammability: Applicable.

Electrical:
Test frequency: 50 MHz to 3 GHz.
Spark test: Not applicable.
Voltage withstanding: 6,500 Vrms, minimum.
Corona extinction voltage: 2,500 Vrms, minimum.
Characteristic impedance: 50 ohms ±2.
Attenuation: See figure 2.
Structural return loss: See figure 3.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross reference of part number.

Part number	Superseded part number or type designation
M17/065-RG165	RG-165/U
M17/065-RG166	RG-166/U

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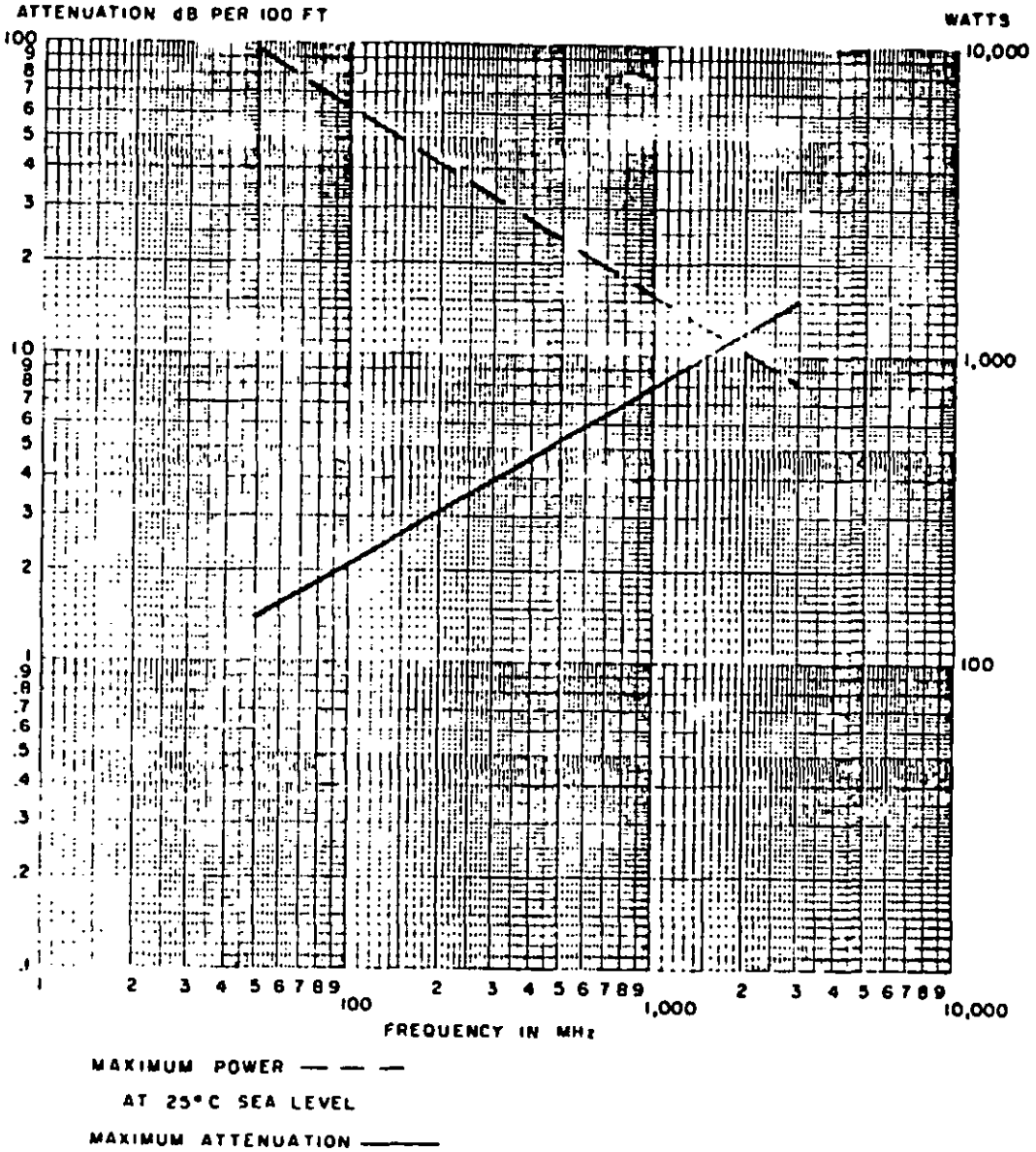


FIGURE 2. Power rating and attenuation.

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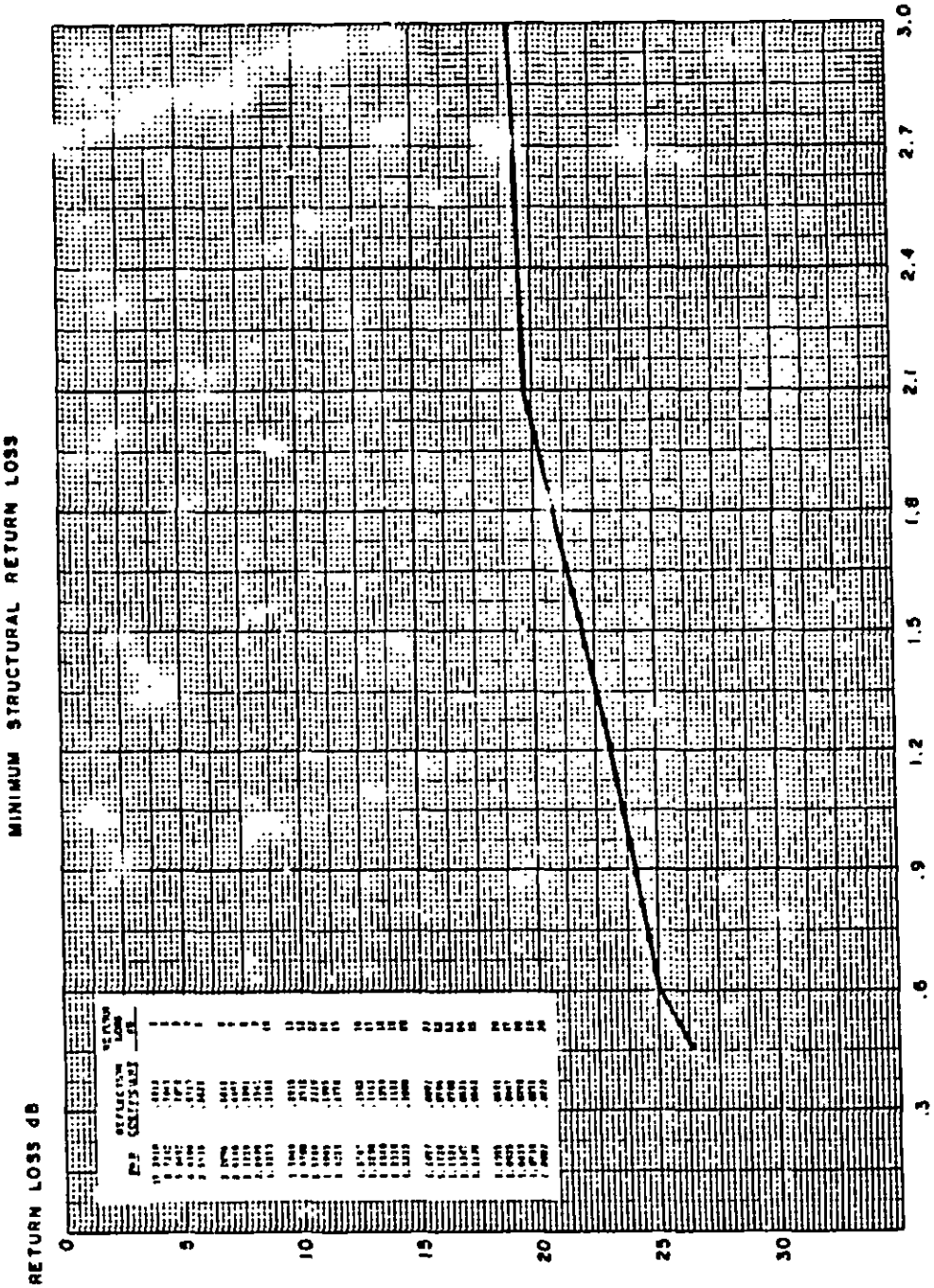


FIGURE 3. Structural return loss.

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Custodians:
Army - EL
Navy - EC
Air Force - 05

Review activities:
Army - EL, MU, MI
Navy - SH, EC
Air Force - 11, 17, 99, 85
DLA - ES

User activities:
Army - ME, AT, SG
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 6145-0689-7)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-17/65B CABLES, RADIO FREQUENCY, FLEXIBLE COAXIAL, 50 OHMS UNARMORED M17/065-RG165, AND ARMORED, M17/065-RG166.

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 58 WHICH MAY BE USED.

ESC-PM 1068-68

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